

# Euclid's Elements — Book XI

## Proposition 17

Formal Reconstruction — Technical Documentation

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*If two straight lines be cut by parallel planes, they shall be cut in the same ratio.*

### List of Figures

|   |                                                      |   |
|---|------------------------------------------------------|---|
| 1 | Three parallel planes cut two transversals . . . . . | 2 |
| 2 | Auxiliary point and the two VI.2 planes . . . . .    | 5 |

## 1 Introduction

Proposition XI.17 is the first proposition in Book XI where the spatial incidence theory and the classical theory of proportion interact in an essential way. Three parallel planes cut two transversals in the orders

$$A - E - B, \quad C - F - D,$$

and Euclid proves

$$AE : EB = CF : FD.$$

The classical construction introduces a point  $O$  on  $AD$  in the middle plane. Proposition XI.16 then produces two pairs of parallel section lines, and Proposition VI.2 is applied in two different planar triangles:

$$AE : EB = AO : OD, \quad AO : OD = CF : FD.$$

Proposition V.11 gives the final equality of ratios.

The formal reconstruction keeps this classical architecture, but it makes three hidden issues explicit.

First, the point  $O$  is not assumed: it is constructed synthetically from the spatial configuration by XI.16 and an ambient Pasch argument. Second, the two applications of VI.2 occur in two explicit induced plane geometries, not in the ambient 3-space. Third, segment proportion is represented by a purely synthetic raw relation based on right-triangle witnesses and angle congruence; no numerical length, segment division, multiplication, real coordinates, or quotient segment arithmetic is used.

One limitation remains visible in the production theorem: planar VI.2 is supplied as an explicit hypothesis

```
hVI2 :
  forall tau : S.Plane,
    HilbertVI2Raw (PlaneGeo Geo tau)
```

rather than derived here from the primitive Hilbert axioms. Thus XI.17 closes the spatial reconstruction around VI.2 without pretending that the whole planar proportion theory has already been reconstructed from first principles.

## 2 Classical Configuration

Let  $\pi_0, \pi_1, \pi_2$  be three parallel planes. Two transversals meet them at

$$A, E, B$$

and

$$C, F, D,$$

respectively, with

$$A - E - B, \quad C - F - D.$$

Thus

$$A, C \in \pi_0, \quad E, F \in \pi_1, \quad B, D \in \pi_2.$$

The conclusion is

$$AE : EB = CF : FD.$$

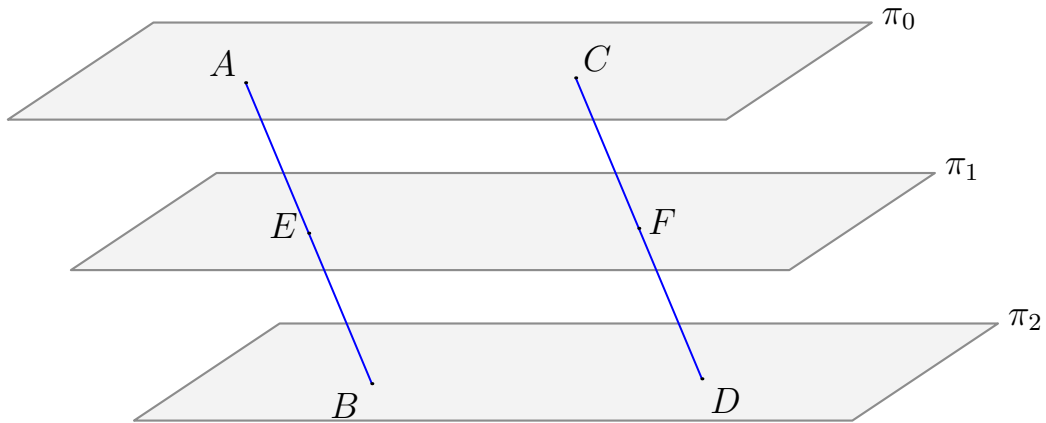


Figure 1: The classical input configuration of Proposition XI.17. The three light planes  $\pi_0, \pi_1, \pi_2$  are pairwise parallel. The two genuinely spatial transversals are shown in blue and meet the planes in the orders  $A - E - B$  and  $C - F - D$ .

The formal statement records pairwise parallelism explicitly:

$$\pi_0 \parallel \pi_1, \quad \pi_1 \parallel \pi_2, \quad \pi_0 \parallel \pi_2.$$

It also records the two betweenness relations directly.

### 3 Statement

**Theorem 1** (Euclid XI.17). *If two straight lines are cut by parallel planes, they are cut in the same ratio.*

In the notation of Figure 1,

$$AE : EB = CF : FD.$$

The current production declaration is:

```

theorem euclid_proposition_11_17
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  (hVI2 :
    forall tau : S.Plane,
      HilbertVI2Raw (PlaneGeo Geo tau))
  (pi0 pi1 pi2 : S.Plane)
  (A E B C F D : Geo.Point)
  (hParallel01 :
    HilbertSpacePlanesParallelIncidence Geo pi0 pi1)
  (hParallel12 :
    HilbertSpacePlanesParallelIncidence Geo pi1 pi2)
  (hParallel02 :
    HilbertSpacePlanesParallelIncidence Geo pi0 pi2)
  (hApi0 : S.OnPlane A pi0)
  (hCpi0 : S.OnPlane C pi0)
  (hEpi1 : S.OnPlane E pi1)
  (hFpi1 : S.OnPlane F pi1)
  (hBpi2 : S.OnPlane B pi2)
  (hDpi2 : S.OnPlane D pi2)
  (hAEB : Geo.Between A E B)
  (hCFD : Geo.Between C F D)
  (hBD : Ne B D)
  (hAC : Ne A C) :
  HilbertSpaceSegmentProportionRaw
    Geo
    A E
    E B
    C F
    F D

```

The hypotheses  $hBD$  and  $hAC$  are explicit nondegeneracy conditions. They guarantee that the two outer-plane connector lines  $BD$  and  $AC$  used in Euclid's construction are genuine lines determined by distinct points.

## 4 Classical Proof

Join  $AC$ ,  $BD$ , and  $AD$ . Let  $AD$  meet the middle plane  $\pi_1$  at  $O$ , and join  $EO$  and  $OF$ .

The plane through  $E, B, D, O$  cuts the two parallel planes  $\pi_1, \pi_2$ . By XI.16 their common sections are parallel:

$$EO \parallel BD.$$

Similarly, the plane through  $A, O, F, C$  cuts the two parallel planes  $\pi_0, \pi_1$ , so XI.16 gives

$$AC \parallel OF.$$

Now work first in triangle  $ABD$ . Since  $E$  lies on  $AB$ ,  $O$  lies on  $AD$ , and

$$EO \parallel BD,$$

VI.2 gives

$$AE : EB = AO : OD.$$

Next work in triangle  $ADC$ . Since  $O$  lies on  $AD$ ,  $F$  lies on  $DC$ , and

$$OF \parallel AC,$$

VI.2 gives

$$AO : OD = CF : FD.$$

By V.11,

$$AE : EB = CF : FD.$$

This is exactly the classical proof pattern recorded in the standard text of XI.17: XI.16 is used twice, VI.2 is used twice, and V.11 composes the two proportions.

**Diagrammatic audit.** The first figure records the input state before Euclid’s construction. The second records the stable proof state after the auxiliary point  $O$  and the two section-parallelisms have been obtained. No separate figure is needed for the raw ratio witnesses: they belong to the formal representation of proportion, not to a new Euclidean configuration.

## 5 The Auxiliary Point $O$

In Euclid’s text the phrase “let  $AD$  meet the plane  $\pi_1$  at  $O$ ” is geometrically natural but formally nontrivial. The production theorem does not take  $O$  as input. It constructs  $O$  in

```
euclid_proposition_11_17_construct_0
```

with conclusion

```
exists O : Geo.Point,
  Geo.Between A O D /\
  S.OnPlane O pi1
```

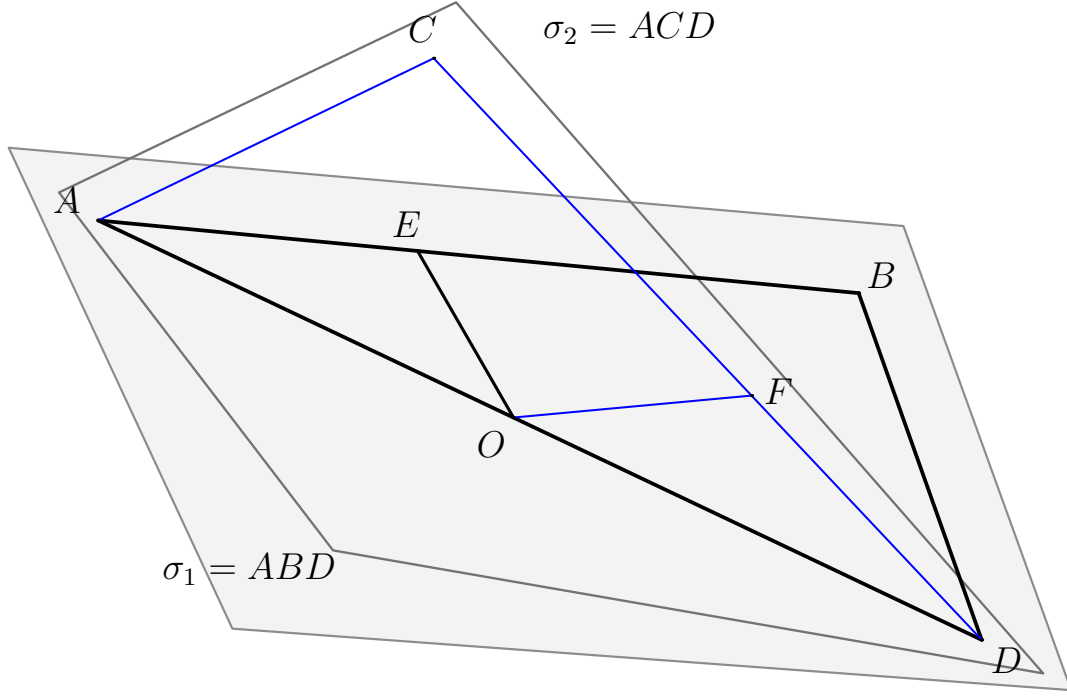


Figure 2: Euclid's auxiliary construction. The reference plane  $\sigma_1 = ABD$  is shown as the main light plane:  $E \in AB$ ,  $O \in AD$ , and  $EO \parallel BD$ . The second plane  $\sigma_2 = ACD$  meets  $\sigma_1$  along  $AD$ ; the genuinely spatial part leaving the reference plane is shown in blue, with  $F \in CD$  and  $OF \parallel AC$ . The two VI.2 applications therefore occur in different planar slices sharing the line  $AD$ .

The construction is synthetic.

First, the proof constructs the plane

$$\sigma = \text{plane}(A, B, D).$$

Because  $A - E - B$ , the point  $E$  also lies in  $\sigma$ .

Next XI.16 is applied to the parallel planes  $\pi_1$  and  $\pi_2$ , cut by  $\sigma$ . It produces the section line through  $E$  and the section line  $BD$ , and proves these two lines parallel. Hence the section through  $E$  is disjoint from  $BD$ .

The line through  $E$  in the middle-plane section meets the open side  $AB$  of triangle  $ABD$ . Ambient Pasch in  $\sigma$  says that it must exit through  $AD$  or  $BD$ . The  $BD$  alternative is impossible because the section line is parallel to  $BD$ . Therefore it meets  $AD$  at a point  $O$ , and the section characterization supplied by XI.16 shows

$$O \in \pi_1.$$

This is one of the main formal gains of the reconstruction: a diagrammatic intersection that Euclid takes as immediate is factored into XI.16 plus the order-theoretic Pasch mechanism.

## 6 What is genuinely three-dimensional here?

XI.17 has two distinct spatial layers.

The first is the original arrangement of three parallel planes and two transversals. The point  $O$  cannot be constructed inside one fixed pre-existing plane; its existence depends on the interaction between the middle plane and the auxiliary plane  $ABD$ .

The second is the construction of the two section-parallelisms

$$EO \parallel BD, \quad AC \parallel OF.$$

Each is produced by XI.16 from a pair of ambient parallel planes and a cutting plane.

Only after these spatial steps are complete does the proof reduce to two planar slices. The first VI.2 calculation is performed in a plane  $\tau_1$  containing  $EO$  and  $BD$ ; the second is performed in a plane  $\tau_2$  containing  $AC$  and  $OF$ . These need not be the same plane.

Thus the formal proof does not “flatten” the whole XI.17 configuration. It moves from ambient 3-space to two different local planar geometries and then transports the resulting proportions back to space.

## 7 Synthetic Representation of Ratio and Proportion

The production proof deliberately avoids segment arithmetic.

A concrete ratio  $PQ : RS$  is represented by a right-triangle witness: one leg is congruent to  $PQ$ , the other to  $RS$ , and the corresponding acute angle represents the ratio. Equality of ratios is then expressed by congruence of the defining acute angles.

The planar relation is

```
HilbertSegmentProportionRaw
```

and its ambient 3D counterpart is

```
HilbertSpaceSegmentProportionRaw
```

The raw VI.2 interface is

```
def HilbertVI2Raw
  [HilbertIncidence Geo]
  [HilbertCongruence Geo] : Prop :=
  forall X E Y F Z : Geo.Point,
    Geo.Between X E Y ->
    Geo.Between X F Z ->
    Geo.Parallel E F Y Z ->
    HilbertSegmentProportionRaw
      Geo X E E Y X F F Z /\
      HilbertSegmentProportionRaw
        Geo E Y X E F Z X F
```

This is exactly the proportional consequence needed from VI.2, stated without division or numerical lengths.

The final V.11-type composition is supplied by transitivity of the raw proportion relation.

## 8 Spatial / Planar Architecture Used

The ambient layer of XI.17 uses

```
HilbertSpacePrimitive  
HilbertSpaceIncidence  
HilbertSpaceOrder  
HilbertSpaceCongruence  
HilbertSpacePlanesParallelIncidence  
HilbertSpaceLinesParallel
```

The key spatial theorem is

```
euclid_proposition_11_16
```

and the auxiliary point construction also uses ambient Pasch through

```
HilbertSpaceOrder.pasch_in_plane
```

The two VI.2 applications live in explicit induced plane geometries

```
PlaneGeo Geo tau1  
PlaneGeo Geo tau2
```

The resulting planar proportions are transported back to ambient space by

```
hilbertSegmentProportionRaw_to_space
```

and finally composed by

```
hilbertSpaceSegmentProportionRaw_trans
```

This respects the permanent Book XI instance discipline: there is no global ambient planar order or planar congruence instance. Planar proportion arguments occur only after an explicit plane slice has been selected.

## 9 Proof Architecture of the Reconstruction

The production proof is organized into five mathematical stages.

### 9.1 Stage 1: construct $O$

The theorem

```
euclid_proposition_11_17_construct_0
```

constructs

$$A - O - D, \quad O \in \pi_1.$$

As explained above, this uses the plane  $ABD$ , XI.16, and Pasch.

## 9.2 Stage 2: obtain the two XI.16 parallelisms

The spatial core constructs four lines:

$$EO, \quad BD, \quad AC, \quad OF$$

and proves

$$EO \parallel BD, \quad AC \parallel OF.$$

The corresponding production helper is

```
euclid_proposition_11_17_spatial_core
```

This stage is purely about ambient incidence, order, and the section theorem XI.16.

## 9.3 Stage 3: first VI.2 slice

From the spatial parallelism  $EO \parallel BD$ , the proof opens a witness plane  $\tau_1$  containing both lines. The points  $A, E, B, O, D$  are placed in the induced geometry  $\text{PlaneGeo}(\tau_1)$ .

The local VI.2 interface then yields

$$AE : EB = AO : OD.$$

This planar proportion is transported back to ambient 3-space.

## 9.4 Stage 4: second VI.2 slice

The parallelism  $AC \parallel OF$  similarly provides a plane  $\tau_2$ . Inside  $\text{PlaneGeo}(\tau_2)$ , the proof uses the reversed betweenness orientations

$$D - O - A, \quad D - F - C$$

to fit the exact argument order of the raw VI.2 interface.

After applying VI.2, the formal proportion is reversed back to the desired orientation by

```
hilbertSegmentProportionRaw_reverse_all
```

and transported to ambient space. The result is

$$AO : OD = CF : FD.$$



## 9.5 Stage 5: V.11

The two ambient raw proportions

$$AE : EB = AO : OD, \quad AO : OD = CF : FD$$

are composed by

```
hilbertSpaceSegmentProportionRaw_trans
```

to obtain

$$AE : EB = CF : FD.$$

## 10 Lean Formalization

The public theorem is intentionally short at the top level. It first calls

```
euclid_proposition_11_17_construct_0
```

to obtain  $O$ , then delegates the complete remaining argument to

```
euclid_proposition_11_17_with_0
```

The latter handles two conceptually different tasks:

- reconstruction of Euclid’s section lines and parallelisms;
- the two plane-local VI.2 calculations and their transport back to ambient space.

This separation is mathematically useful. It isolates the genuinely spatial existence problem for  $O$  from the proportion calculation that begins once  $O$  is available.

## 11 Hidden Formal Data

Several pieces of information visible in Euclid’s diagram become explicit proof obligations.

- Pairwise parallelism of the three planes is given explicitly.
- The order relations  $A - E - B$  and  $C - F - D$  are hypotheses, not inferred from a picture.
- The connector points  $A, C$  and  $B, D$  are required to be distinct:  $A \neq C$  and  $B \neq D$ .
- The point  $O$  must be proved to exist, to satisfy  $A - O - D$ , and to lie in the middle plane.
- The planes  $ABD$  and  $ACD$  must be constructed from verified noncollinearity.
- XI.16 supplies section lines pointwise; their incidences with  $E, O, B, D, A, C, F$  must be recovered from those characterizations.

- Each spatial line-parallelism carries a witness plane, and that plane becomes the correct **PlaneGeo** slice for VI.2.
- The second VI.2 application requires orientation normalization before the resulting proportion can be put into Euclid’s final order.

These obligations explain why the formal DAG is longer than Euclid’s visible proof even though the mathematical strategy is the same.

## 12 Relationship with Earlier Book XI Propositions

XI.16 is the decisive Book XI dependency, both classically and formally.

Classically it is invoked twice:

$$EO \parallel BD, \quad AC \parallel OF.$$

Formally it plays an additional role in the construction of  $O$ : the middle and lower parallel planes are cut by the plane  $ABD$ , and the resulting section line through  $E$  is shown parallel to  $BD$ . This disjointness is what eliminates one of the two Pasch exits.

Thus the formal use of XI.16 is slightly richer than the visible textbook proof:

- once to construct  $O$ ;
- twice to obtain the parallel section pairs used in the ratio proof.

No Wyler flat calculus is imported. XI.17 depends on the canonical synthetic XI.16 in the main Book XI route.

## 13 Relationship with Books V and VI / Hilbert Infrastructure

The classical proportion dependencies are exactly:

$$\text{VI.2}, \quad \text{V.11}.$$

In the current formal architecture they appear in synthetic raw form.

VI.2 is represented by the explicit plane-local assumption

```
forall tau : S.Plane,
  HilbertVI2Raw (PlaneGeo Geo tau)
```

and V.11 by transitivity of

```
HilbertSpaceSegmentProportionRaw
```

The right-triangle representation of ratio uses Hilbert congruence and right-angle infrastructure. The production audit also confirms that no new XI.17-specific geometric axiom was introduced.

However, because VI.2 is an explicit hypothesis, the theorem should not yet be described as a derivation of Euclidean proportion theory from primitive Hilbert incidence, order, and congruence alone.

No Group IV typeclass appears in the theorem. The spatial section geometry itself remains independent of the Euclidean parallel postulate.

## 14 Dependency Summary

**Classical.**

$$\begin{array}{c}
\pi_0 \parallel \pi_1 \parallel \pi_2, \quad A - E - B, \quad C - F - D \\
\downarrow \\
\text{join } AC, BD, AD; \quad O = AD \cap \pi_1 \\
\downarrow \\
\text{XI.16 : } EO \parallel BD, \quad AC \parallel OF \\
\downarrow \\
\text{VI.2 in } \triangle ABD : AE : EB = AO : OD \\
\downarrow \\
\text{VI.2 in } \triangle ADC : AO : OD = CF : FD \\
\downarrow \\
\text{V.11} \\
\downarrow \\
AE : EB = CF : FD.
\end{array}$$

**Formal.**

```

Hilbert 3D incidence/order/congruence
+ HilbertVI2Raw on every PlaneGeo
|
v
Proposition11_16 + ambient Pasch
|
v
euclid_proposition_11_17_construct_0
|
v
euclid_proposition_11_17_spatial_core
|
v
PlaneGeo tau1: AE:EB = AO:OD
|
v
PlaneGeo tau2: AO:OD = CF:FD
|
v
transport to ambient raw proportions
|
v
hilbertSpaceSegmentProportionRaw_trans
|
v
euclid_proposition_11_17

```

The main conceptual difference is the treatment of  $O$ . Euclid names the intersection directly. Lean proves its existence from the spatial incidence/order infrastructure before the classical VI.2 chain can begin.

## 15 Audit Status

This documentation has been checked against the current production theorem `euclid_proposition_11_17` and the current raw proportion module. The local production build and the combined XI.16–XI.17 audit were reported clean.

The audit is consistent with the intended architecture: XI.17 introduces no new proposition-specific geometric axiom and does not use segment arithmetic or numerical coordinates. Its remaining explicit mathematical interface is the plane-local hypothesis `HilbertVI2Raw`.

The two figures represent the two stable geometric states used in the documentation: the original three-plane transversal configuration and the post-construction state containing  $O$  and the two VI.2 slices.

## 16 Conclusion

Proposition XI.17 is the point where the Book XI reconstruction begins to combine genuinely spatial geometry with planar proportion theory.

Its 3D content is now cleanly synthetic. XI.16 supplies the section parallelisms, and XI.16 together with Pasch constructs the auxiliary point  $O$  that Euclid introduces without further comment. The ratio calculation then takes place in two explicit planar slices and is transported back to ambient space.

The result is therefore a useful separation theorem for the architecture of the project. Spatial incidence and order are handled in the Book XI layer; ratio and proportion are handled by a separate synthetic interface. What remains for a deeper foundational reconstruction is not the 3D geometry of XI.17, but the derivation of the raw VI.2 interface itself.